



INSTRUCTOR SOLUTION

Habib University
Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026
Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes

Quiz No 05

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO Assessed – CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3) – (applicable to both the questions.)

Question 1 [10 points] [Joint Distribution & Independence]

The joint probability distribution for the two Discrete Random Variables X and Y is shown below. The probability of the event $P(Y > X) = \frac{9}{50}$

Part a [5 points] Find the values for a & b.

Part b [5 points] Determine (with justification) whether random Variable X and Y are independent or not.

Y=	3	$\frac{1}{25}$	a	b
	2	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{5}{25}$
	1	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{5}{25}$
		1	2	3
		X=		

1 (a) $P(Y > X) = 9/50$

$$\Rightarrow \sum_{\substack{x,y: \\ y > x}} P_{X,Y}(x,y) = 9/50$$

$$\sum_{\substack{(1,2), (1,3), \\ (2,3)}} P_{X,Y}(x,y) = 9/50 \quad \text{--- (1)}$$

$$\sum_{x,y} P_{X,Y}(1,2) + P_{X,Y}(1,3) + P_{X,Y}(2,3) = 9/50$$

$$\frac{2}{25} + \frac{1}{25} + a = 9/50 \quad \text{--- (1)}$$

$$a = \frac{9}{50} - \frac{3}{25} = \frac{3}{50} = \frac{1.5}{25} \quad \text{--- (1)}$$

Also, $\sum_{(x,y)} P_{X,Y}(x,y) = 1$

Joint & Valid
Pmf.

$\Rightarrow$ Summing the 9 Probabilities (Joint)

$$= \left(\frac{2}{25} + \frac{2}{25} + \frac{1}{25}\right) + \left(\frac{3}{25} + \frac{3}{25} + a\right) + \left(\frac{5}{25} + \frac{5}{25} + b\right) = 1$$

$$a + b = 1 - \frac{21}{25} = \frac{4}{25}$$

$$b = \frac{4}{25} - a = \frac{4}{25} - \frac{1.5}{25}$$

$$= \frac{2.5}{25}$$

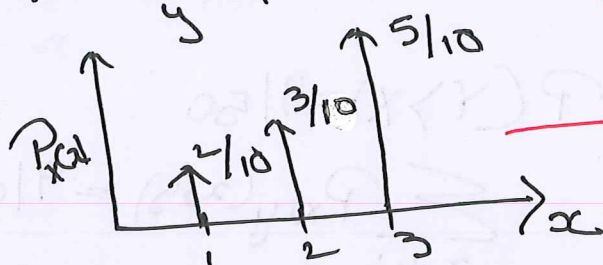
(b) For R.V X & Y.

to be independent, we need to show
 $P_{X,Y}(x,y) = P_X(x) \cdot P_Y(y)$ for all x & y .

Lets find $P_X(x)$; $P_X(x) = \sum_y P_{X,Y}(x,y)$

$$P_X(1) = \frac{5}{25}, P_X(2) = \frac{7.5}{25}$$

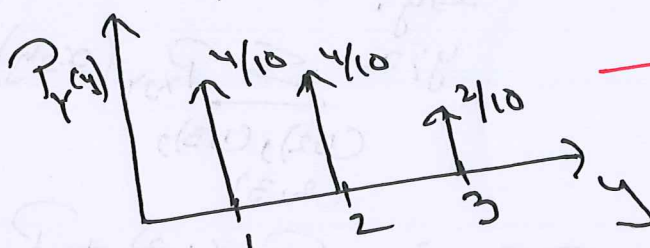
$$P_X(3) = \frac{12.5}{25}$$



$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$P_Y(1) = \frac{10}{25}, P_Y(2) = \frac{10}{25}$$

$$P_Y(3) = \frac{5}{25}$$



Lets test (A)
we see that for all 9 combinations of x, y

$$P_{X,Y}(x,y) = P_X(x) \cdot P_Y(y)$$

Hence X & Y ARE INDEPENDENT.

Question 2 [10 points] [Function of a Random Variable & Expectation of a R.V.]

We have a tetrahedral die (4-sided) that has four outcomes, each side bearing numbers 1, 2, 3 and 4 with equal probabilities (fair die). We define X_1 and X_2 as the first and second outcome of the two independent rolls of the tetrahedral die, respectively. We define the following Discrete Random Variable:

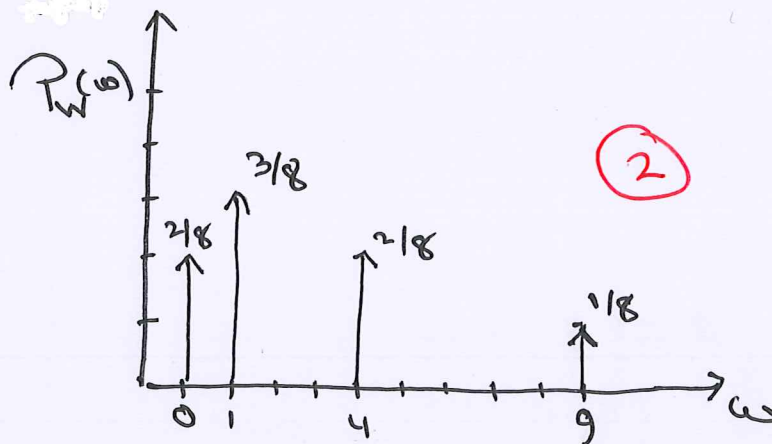
$$W = (X_1 - X_2)^2$$

- (a) [4 points] Calculate and plot the Probability Mass Function, for the DRV, W .
 (b) [6 points] Calculate the Expected value, $E[W]$ and $\text{Var}[W]$.

2(a)

X_1	X_2	Probabilities	$W = (X_1 - X_2)^2$
1	1	1/16	0
1	2	1/16	1
1	3	1/16	4
1	4	1/16	9
2	1	1/16	1
2	2	1/16	0
2	3	1/16	1
2	4	1/16	4
3	1	1/16	4
3	2	1/16	1
3	3	1/16	0
3	4	1/16	1
4	1	1/16	9
4	2	1/16	4
4	3	1/16	1
4	4	1/16	0

$W =$	Occurrences	Probability
0	4	$4/16 = 2/8$
1	6	$6/16 = 3/8$
4	4	$4/16 = 2/8$
9	2	$2/16 = 1/8$



$$(b) \quad E[W] = \sum_w w P_W(w) = 0 \cdot \frac{2}{8} + 1 \cdot \frac{3}{8} + 4 \cdot \frac{2}{8} + 9 \cdot \frac{1}{8} = 2.5$$

$$\text{Var}[W] = E[W^2] - (E[W])^2$$

$$\hookrightarrow E[W^2] = \sum_w w^2 P_W(w) = 0 \times \frac{2}{8} + 1 \times \frac{3}{8} + 16 \times \frac{2}{8} + 81 \cdot \frac{1}{8}$$

$$= 14.5$$

$$\text{Var}[W] = 14.5 - (2.5)^2 = 14.5 - 6.25$$

$$= 8.25$$